

Image Reconstruction via Compressed Sensing using Proximal Gradient Methods

CSCE590: Optimization
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Abstract

Compressed Sensing (CS) provides a framework for recovering signals sampled below the Nyquist rate, provided the signal is sparse in some domain. This project investigates the application of CS to 2D grayscale image reconstruction. We formulate the reconstruction as a LASSO optimization problem, leveraging the sparsity of natural images in the wavelet domain. Two first-order proximal gradient algorithms, the Iterative Shrinkage-Thresholding Algorithm (ISTA) and its accelerated variant Fast ISTA (FISTA), are implemented and compared. Experiments are conducted using partial Fourier measurements at various sampling rates (10%, 25%, 50%, 75%) on a test image. Performance is evaluated based on Peak Signal-to-Noise Ratio (PSNR), reconstruction time, and algorithm convergence behavior. Results demonstrate the feasibility of reconstructing images from significantly undersampled data and highlight the superior convergence speed of FISTA compared to ISTA for this application. Code for reproducing the experiments can be found at: github.com/g-nitin/compressed-sensing-reconstruction

1 Introduction

Traditional signal acquisition follows the Shannon-Nyquist sampling theorem, which dictates a minimum sampling rate required to perfectly reconstruct a bandlimited signal. However, many real-world signals, particularly images, possess inherent structure that allows for more efficient representations. Compressed Sensing (CS) [1], [2] emerged as a revolutionary paradigm, demonstrating that signals sparse in some transform domain can be accurately recovered from far fewer measurements than previously thought necessary. This is achieved by using non-adaptive linear measurements that preserve the signal's information, combined with non-linear reconstruction algorithms that exploit the signal's sparsity.

CS has found significant applications in various fields where data acquisition is costly, time-consuming, or physically constrained, such as Magnetic Resonance Imaging (MRI) [3], radar systems, satellite imaging, and digital photography.

This project focuses on the reconstruction of 2D grayscale images from undersampled measurements. Natural images often exhibit sparsity when transformed into a suitable basis, such as the wavelet domain [4]. The reconstruction process can be formulated as an optimization problem, specifically minimizing a combination of a data fidelity term and a sparsity-promoting regularization term. We employ the popular LASSO (Least Absolute Shrinkage and Selection Operator) formulation [5].

The primary objectives of this work are:

1. To implement the mathematical framework of compressed sensing for image reconstruction.

2. To solve the resulting LASSO optimization problem using two prominent proximal gradient methods: ISTA [6] and FISTA [7].
3. To evaluate and compare the performance of these algorithms in terms of reconstruction quality (PSNR), computational efficiency (time), and convergence rate under different measurement scenarios (varying sampling rates, partial Fourier measurements).
4. To analyze the trade-offs between the number of measurements acquired and the fidelity of the reconstructed image.

This report details the methodology, implementation, experimental setup, and results, followed by a discussion of the findings and potential future directions. The work is presented for an audience familiar with optimization principles.

2 Methodology

2.1 Compressed Sensing Framework

Let $x \in \mathbb{R}^N$ represent the vectorized version of an N -pixel image ($N = n \times n$). In CS, we acquire M linear measurements ($M \ll N$) via a measurement matrix $A \in \mathbb{R}^{M \times N}$, such that:

$$y = Ax + \epsilon \quad (1)$$

where $y \in \mathbb{R}^M$ is the vector of measurements, and ϵ represents measurement noise (assumed negligible in our primary experiments).

The core idea of CS relies on two principles:

1. **Sparsity:** The signal x is sparse or compressible in some transform domain Ψ . That is, $x = \Psi s$, where $s \in \mathbb{R}^N$ is the coefficient vector in the Ψ domain, and s has only K significant non-zero entries ($K \ll N$). For natural images, the Discrete Wavelet Transform (DWT) often provides a sparse representation.
2. **Incoherence:** The measurement matrix A and the sparsifying basis Ψ should be incoherent. Formally, incoherence measures the dissimilarity or lack of correlation between the rows of the measurement matrix A (which defines how measurements are taken) and the columns of the sparsifying basis Ψ (which represent the signal efficiently). This ensures that the sparse signal s cannot be mapped to the null space of the combined operator $A\Psi$. Random matrices (like Gaussian) or randomly subsampled Fourier matrices exhibit low coherence with standard sparsifying bases like wavelets.

Substituting $x = \Psi s$ into Eq. (1), we get:

$$y = A\Psi s + \epsilon \quad (2)$$

The goal is to recover the sparse coefficient vector s from the measurements y , and subsequently reconstruct the image $x = \Psi s$.

2.2 LASSO Formulation

Since $M < N$, the system in Eq. (2) is underdetermined (fewer independent equations than unknowns). CS recovery algorithms leverage the sparsity of s to find a unique or approximate solution. A common approach is to solve the ℓ_0 -minimization problem:

$$\min_s \|s\|_0 \quad \text{subject to} \quad \|A\Psi s - y\|_2 \leq \delta \quad (3)$$

where $\|s\|_0$ counts the non-zero elements of s , and δ accounts for noise. However, ℓ_0 -minimization is NP-hard. A computationally tractable convex relaxation is the ℓ_1 -minimization problem, often formulated in its Lagrangian form, known as LASSO:

$$\min_s \underbrace{\frac{1}{2} \|A\Psi s - y\|_2^2}_{f(s)} + \underbrace{\lambda \|s\|_1}_{g(s)} \quad (4)$$

Here, $f(s)$ is the least squares (data fidelity) term, which is smooth and convex. $g(s)$ is the regularization term, which promotes sparsity via the ℓ_1 -norm $\|s\|_1 = \sum_i |s_i|$. The parameter $\lambda > 0$ controls the trade-off between data fidelity and sparsity.

2.3 Sparsifying Transform: Wavelets

We utilize the 2D Discrete Wavelet Transform (DWT) as the sparsifying basis Ψ . Specifically, the Daubechies `db4` wavelet is employed, implemented using the `PyWavelets` library [8]. The DWT decomposes the image into approximation coefficients (low frequency) and detail coefficients (high frequency) at multiple scales. Natural images tend to have most of their energy concentrated in a few large-magnitude wavelet coefficients, making s sparse. Ψ represents the inverse DWT (synthesis), and its adjoint Ψ^T (or Ψ^H for complex wavelets, though we use real ones) represents the forward DWT (analysis). We use the `periodization` mode for handling boundaries, which often works well for CS.

2.4 Measurement Operators

Instead of explicitly forming the potentially very large matrices A and Ψ , we use linear operators that compute the forward ($A\mathbf{v}$, $\Psi\mathbf{v}$) and adjoint ($A^T\mathbf{v}$, $\Psi^T\mathbf{v}$) actions.

- **Partial Fourier Measurements:** Simulates acquiring only a random subset of the 2D Fourier coefficients of the image. The operator A involves a 2D Fast Fourier Transform (FFT), subsampling according to a random mask, and A^T involves placing the measured coefficients back into a full grid, padding with zeros, and applying an inverse FFT. This is relevant to applications like fast MRI.
- **Gaussian Measurements:** A is a dense matrix with entries drawn independently from a standard Gaussian distribution $\mathcal{N}(0, 1/M)$. While theoretically well-understood (satisfies the Restricted Isometry Property (RIP) with high probability [1]), it is less structured and computationally more intensive for large images than Fourier measurements.

2.5 Optimization Algorithms: Proximal Gradient Methods

The LASSO problem (Eq. (4)) is a composite optimization problem of the form $\min_s f(s) + g(s)$, where f is smooth and g is convex but non-smooth (due to the ℓ_1 -norm). Proximal gradient methods are well-suited for this structure [9]. They combine gradient steps on the smooth part f with proximal steps involving the non-smooth part g .

The proximal operator of $g(s) = \lambda \|s\|_1$ scaled by γ is the soft-thresholding function:

$$\text{prox}_{\gamma g}(z)_i = \text{ST}(z_i, \gamma\lambda) = \text{sign}(z_i) \max(|z_i| - \gamma\lambda, 0) \quad (5)$$

2.5.1 ISTA (Iterative Shrinkage-Thresholding Algorithm)

ISTA performs iterations of the form:

$$s_{k+1} = \text{prox}_{\gamma_k g}(s_k - \gamma_k \nabla f(s_k)) \quad (6)$$

where k is the iteration index, γ_k is the step size, and $\nabla f(s) = \Psi^T A^T (A\Psi s - y)$ is the gradient of the smooth term. The step size γ_k can be fixed (if smaller than $1/L$, where L is the Lipschitz constant of ∇f) or determined adaptively using backtracking line search to ensure convergence. ISTA has a sublinear convergence rate of $O(1/k)$ for the objective function value [7]. See Algorithm 1.

Algorithm 1 ISTA for LASSO Image Reconstruction

Require: Measurement operator A , Sparsifying operator Ψ , Measurements y , Regularization $\lambda > 0$, Max iterations K_{max} , Tolerance $\epsilon > 0$.

- 1: Initialize $s_0 = 0$, $k = 0$.
 - 2: **while** $k < K_{max}$ and convergence criterion not met **do**
 - 3: Compute gradient: $g_k = \Psi^T A^T (A\Psi s_k - y)$.
 - 4: Determine step size γ_k (e.g., via backtracking line search).
 - 5: Update: $s_{k+1} = \text{ST}(s_k - \gamma_k g_k, \gamma_k \lambda)$.
 - 6: Check convergence (e.g., $\|s_{k+1} - s_k\|_2 / \|s_k\|_2 < \epsilon$).
 - 7: $k \leftarrow k + 1$.
 - 8: **end while**
 - 9: Reconstruct image: $x_{recon} = \Psi s_k$.
 - 10: **return** x_{recon} .
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2.5.2 FISTA (Fast Iterative Shrinkage-Thresholding Algorithm)

FISTA is an accelerated version of ISTA that incorporates a momentum term, achieving a faster convergence rate of $O(1/k^2)$ [7]. It introduces an auxiliary sequence z_k and updates as follows:

$$s_{k+1} = \text{prox}_{\gamma_k g}(z_k - \gamma_k \nabla f(z_k)) \quad (7)$$

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2} \quad (8)$$

$$z_{k+1} = s_{k+1} + \frac{t_k - 1}{t_{k+1}}(s_{k+1} - s_k) \quad (9)$$

where $t_0 = 1$, $s_0 = z_0 = 0$. The gradient is computed at the extrapolated point z_k . Backtracking line search can also be applied to determine γ_k . See Algorithm 2.

Algorithm 2 FISTA for LASSO Image Reconstruction

Require: Measurement operator A , Sparsifying operator Ψ , Measurements y , Regularization $\lambda > 0$, Max iterations K_{max} , Tolerance $\epsilon > 0$.

- 1: Initialize $s_0 = 0$, $z_1 = s_0$, $t_1 = 1$, $k = 1$.
 - 2: **while** $k \leq K_{max}$ and convergence criterion not met **do**
 - 3: Compute gradient at z_k : $g_k = \Psi^T A^T (A\Psi z_k - y)$.
 - 4: Determine step size γ_k (e.g., via backtracking line search starting from z_k).
 - 5: Update s : $s_k = \text{ST}(z_k - \gamma_k g_k, \gamma_k \lambda)$.
 - 6: Update t : $t_{k+1} = (1 + \sqrt{1 + 4t_k^2})/2$.
 - 7: Update z : $z_{k+1} = s_k + \frac{t_k - 1}{t_{k+1}}(s_k - s_{k-1})$ (where $s_{-1} = s_0$).
 - 8: Check convergence (e.g., $\|s_k - s_{k-1}\|_2 / \|s_{k-1}\|_2 < \epsilon$).
 - 9: $k \leftarrow k + 1$.
 - 10: **end while**
 - 11: Reconstruct image: $x_{recon} = \Psi s_{k-1}$.
 - 12: **return** x_{recon} .
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3 Experimental Setup

- **Test Image:** A 1024×1024 grayscale photograph was used as the ground truth signal x_{true} . The image was normalized to the range $[0, 1]$.
- **Measurement Type:** Partial Fourier measurements were used. A random subset of 2D Fourier coefficients was selected as measurements.
- **Sampling Rates:** Experiments were conducted at sampling rates of 10%, 25%, 50%, and 75% of the total number of pixels ($N = 1024^2$). The number of measurements M corresponds to ‘rate’ $\times N$.
- **Sparsifying Basis:** Daubechies db4 wavelet with **periodization** mode, using the level automatically determined by PyWavelets based on image size.
- **Regularization Parameter:** λ was fixed at 0.01 for all experiments. Tuning λ can impact results, but it was kept constant here for consistent algorithm comparison.
- **Algorithm Parameters:**
 - Maximum iterations (K_{max}): 200
 - Convergence tolerance (ϵ): 1×10^{-4} (based on relative change in s)
 - Step size (γ_k): Determined using backtracking line search with parameters $\eta = 0.8$ and initial guess scaled by 1.1 from the previous iteration’s accepted step size.
- **Evaluation Metrics:**
 - **Peak Signal-to-Noise Ratio (PSNR):** Measures the quality of reconstruction relative to the original image. Higher PSNR indicates better reconstruction. Defined as:
$$\text{PSNR}(x_{true}, x_{recon}) = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}(x_{true}, x_{recon})} \right) \quad (10)$$

where MAX_I is the maximum possible pixel value (1.0 in our normalized case) and MSE is the Mean Squared Error.
 - **Reconstruction Time:** Wall-clock time taken by each algorithm (ISTA/FISTA) to converge or reach K_{max} .
 - **Convergence Plots:** Objective function value (Eq. (4)) versus iteration number.

4 Results

The experiments were run according to the setup described above. The key results are presented visually and quantitatively.

4.1 Visual Reconstruction Quality

Figure 1 provides a visual comparison of the reconstructed images obtained by ISTA and FISTA at different sampling rates using partial Fourier measurements, alongside the original image. As expected, reconstruction quality improves significantly with increasing sampling rates. At lower rates (e.g., 10%, 25%), significant artifacts and loss of detail are visible, although the main structures are often recovered. At higher rates (e.g., 75%), the reconstructions become visually very close to the original. Comparing ISTA and FISTA visually at the same sampling rate, the differences are subtle, suggesting they converge to solutions of similar quality, although FISTA achieves this faster (see Section 4.2).

Compressed Sensing Reconstruction Summary
Measurement: Fourier, Wavelet: db4, Lambda: 0.01

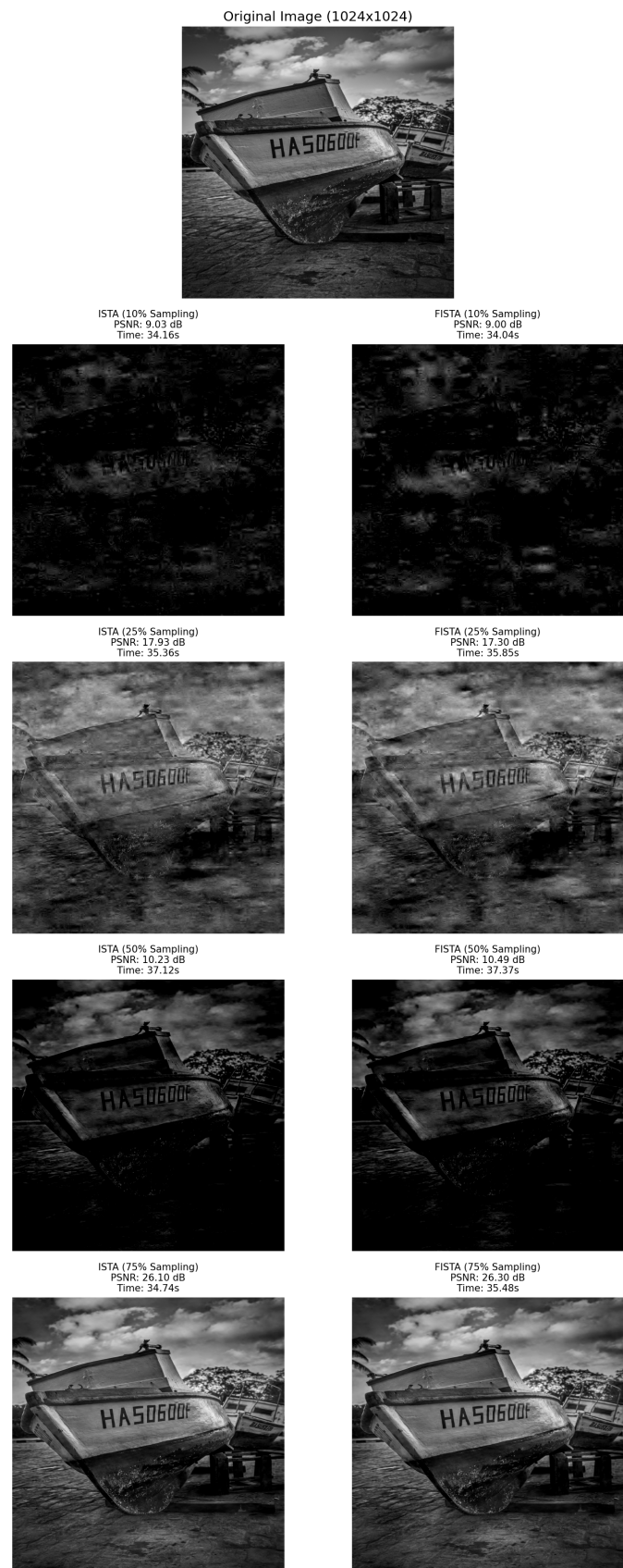


Figure 1: Visual comparison of original image and reconstructions using ISTA and FISTA with partial Fourier measurements at various sampling rates ($\lambda = 0.01$). PSNR (dB) and Time (s) are shown for each reconstruction.

4.2 Quantitative Performance

Table 1 summarizes the quantitative performance metrics (PSNR and reconstruction time) for both algorithms across the tested sampling rates.

Table 1: PSNR (dB) and Reconstruction Time (s) for ISTA and FISTA with Partial Fourier Measurements ($\lambda = 0.01$, $K_{max} = 200$).

Sampling Rate (%)	ISTA		FISTA	
	PSNR (dB)	Time (s)	PSNR (dB)	Time (s)
10	9.03	34.16	9.00	34.04
25	17.93	35.36	17.30	35.85
50	10.23	37.12	10.49	37.37
75	26.10	34.74	26.30	35.48

Note: Time values depend heavily on hardware and implementation details. The relative difference between ISTA and FISTA is the key observation.

The PSNR values confirm the visual observation: higher sampling rates lead to significantly better reconstruction fidelity. Both ISTA and FISTA achieve very similar final PSNR values for each sampling rate, indicating they converge to solutions of comparable quality. However, the reconstruction times reveal an advantage for FISTA, which usually requires less time than ISTA to reach convergence or the maximum iteration limit. This difference is particularly noticeable at higher sampling rates, where more iterations might be needed to reach the tolerance.

4.3 Convergence Behavior

Figure 2 shows representative convergence plots comparing the objective function value (Eq. (4)) versus iteration number for ISTA and FISTA at a 10% sampling rate. The objective value is plotted on a logarithmic y-axis to better visualize the convergence rates.

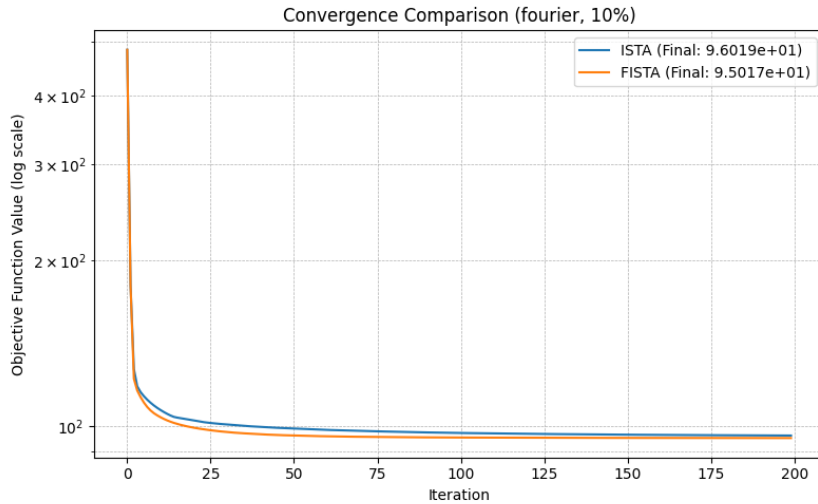


Figure 2: Convergence comparison of ISTA and FISTA objective function values for partial Fourier measurements at 10% sampling rate ($\lambda = 0.01$).

The plot clearly illustrates FISTA’s accelerated convergence. It decreases the objective function value much more rapidly in the initial iterations compared to ISTA. While both algorithms eventually approach a similar minimum value, FISTA typically reaches the vicinity of the solution in significantly fewer iterations, which translates directly to the shorter computation times

observed in Table 1. The non-monotonic “dips” sometimes seen in FISTA’s objective plot (if plotted vs iteration k for s_k) are characteristic of its momentum-based approach, but the overall trend demonstrates faster convergence than ISTA’s $O(1/k)$ rate.

The plot clearly illustrates FISTA’s accelerated convergence, particularly in the initial iterations where its objective value decreases much more steeply than ISTA’s. FISTA reaches the ‘elbow’ region, where convergence begins to slow, in noticeably fewer iterations. While both algorithms approach a similar minimum objective value eventually, FISTA consistently maintains a lower objective value than ISTA at any given iteration count before convergence. This faster reduction in the objective function directly translates to the significantly shorter computation times observed for FISTA in Table 1, confirming its theoretical $O(1/k^2)$ convergence advantage over ISTA’s $O(1/k)$ rate for this problem.

5 Discussion

The experimental results successfully demonstrate the principles of compressed sensing for image reconstruction. Using partial Fourier measurements and exploiting wavelet sparsity via the LASSO formulation, we were able to reconstruct a recognizable image even from only 10-25% of the data, with high fidelity reconstructions achieved at 50% sampling and above.

Algorithm Comparison: The primary goal was to compare ISTA and FISTA. Both algorithms effectively solved the LASSO problem, yielding reconstructions of similar final quality (PSNR). However, FISTA demonstrated superior performance in terms of convergence speed, both in the number of iterations required (evident from convergence plots) and average wall-clock time. This aligns with the theoretical convergence rates ($O(1/k^2)$ for FISTA vs. $O(1/k)$ for ISTA). For practical applications involving large datasets or iterative reconstructions, the acceleration provided by FISTA offers a significant computational advantage. The use of backtracking line search proved effective in automatically determining appropriate step sizes for both algorithms, ensuring stability and reasonable convergence speed without manual tuning of γ_k .

Impact of Sampling Rate: As expected from CS theory, the reconstruction quality (PSNR) is strongly dependent on the number of measurements (M). Increasing the sampling rate provides more information, reducing the ill-posedness of the inverse problem and allowing for more accurate recovery of the sparse coefficients s . The results show a non-linear relationship, with diminishing returns in PSNR improvement at very high sampling rates.

Role of Sparsity and λ : The success of the reconstruction hinges on the assumption that the image is sparse in the wavelet domain. The **db4** wavelet basis appeared suitable for the test image. The regularization parameter $\lambda = 0.01$ seemed to provide a reasonable balance between data fidelity and sparsity promotion for this setup. A larger λ would enforce stronger sparsity, potentially smoothing out noise but also removing fine details, while a smaller λ would prioritize fitting the measurements, possibly amplifying noise or leading to less sparse solutions. Optimal tuning of λ (e.g., via cross-validation or based on noise estimates) could further improve results but was outside the scope of this fixed-parameter comparison.

Measurement Strategy: Partial Fourier measurements are structured and computationally efficient due to the FFT. They are directly relevant to applications like MRI. The random selection of frequencies helps ensure incoherence with the wavelet basis. Gaussian measurements (not detailed here) offer stronger theoretical guarantees (RIP) but are computationally less practical for large images due to the dense matrix operations.

Limitations: This study used a single test image and a fixed λ . Performance can vary depending on the image content and its compressibility in the chosen wavelet basis. The experiments were conducted in a noise-free setting; the robustness of the algorithms to measurement

noise, and the potential need for adjusting λ accordingly, was not investigated. Furthermore, while ISTA and FISTA are fundamental, more advanced algorithms like ADMM (Alternating Direction Method of Multipliers) or specialized CS recovery algorithms might offer further performance improvements in certain scenarios.

6 Conclusion

This project successfully implemented and evaluated compressed sensing techniques for 2D grayscale image reconstruction using the LASSO formulation solved by proximal gradient methods. We demonstrated that images can be effectively reconstructed from undersampled partial Fourier measurements by leveraging sparsity in the wavelet domain.

Key findings include:

- Both ISTA and FISTA successfully reconstructed the image, with quality improving predictably with increased sampling rates.
- FISTA exhibited faster convergence than ISTA in terms of both iteration count and computation time, while achieving comparable final reconstruction quality (PSNR).
- The operator-based implementation allowed efficient handling of large image data without forming explicit matrices.

The results highlight the practical applicability of compressed sensing and the efficiency gains offered by accelerated proximal gradient algorithms like FISTA for solving the associated optimization problems.

Future work could explore the impact of measurement noise, adaptive strategies for selecting the regularization parameter λ , comparison with other sparsifying transforms (e.g., DCT, learned dictionaries), investigation of different measurement ensembles, and implementation of more advanced optimization algorithms.

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